TOOLS RENTAL MANAGEMENT

Odoo Module | Technical & Functional Documentation

Version 19.0.2.1.0

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Category: Rental | Odoo 19+

1. Module Overview

Tools Rental Management is a comprehensive Odoo module designed for businesses that rent tools and equipment. It provides an end-to-end rental workflow — from creating orders and checking out equipment to tracking returns, managing expenses, and generating invoices — all within a single, integrated solution.

Module Name: Tools Rental Management
Technical Name: tools_rental_management
Odoo Version: 19.0+
Version: 19.0.2.1.0
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1.1 Key Features

- Tool & Equipment catalog with real-time availability tracking
- Flexible rental pricing — Daily, Weekly, and Monthly billing periods
- Check-Out and Check-In workflow with photo capture and signatures
- Automatic cost calculation based on duration and pricing rules
- Late return penalty configuration and automatic late-fee calculation
- Partial return support for multi-unit rental orders
- Advance payment collection at checkout
- Damage charge tracking and discount authorization workflow
- Customer rating system at check-in
- Tax wizard for applying GST/VAT on rental lines
- Standalone expense management with category configuration
- Dashboard with live KPIs and analytics
- PDF Rental Agreement, Checkout Invoice, and Check-In Invoice reports
- Multi-company support with company-level access rules
- Full chatter / activity integration on orders and tools

1.2 Module Dependencies

The module requires the following Odoo core apps:

Dependency	Purpose
base	Core Odoo framework, companies, currencies
product	Product catalog, linked product records
stock	Inventory tracking and product availability

purchase	Vendor integration for tool procurement
contacts	Customer partner records
mail	Chatter, activity tracking, email threading

2. Data Models

The module defines the following core data models (database tables). Each model represents a business entity with its own set of fields, constraints, and computed values.

2.1 rental.tool — Tool / Equipment

Represents a single physical tool or piece of equipment available for rent. Each tool has its own identity, category, status, pricing rules, and availability tracking.

Field	Type	Description
name	Char	Tool name (required)
code	Char	Internal reference / SKU
category_id	Many2one	Link to rental.tool.category (required)
state	Selection	Available / Rented / Under Maintenance / Retired
serial_number	Char	Serial number for physical identification
barcode	Char	Unique barcode (scan support); must be unique
total_qty	Float	Total units of this tool owned
available_qty	Float (computed)	total_qty minus quantity currently checked out
pricing_ids	One2many	Pricing rules (rental.pricing) for this tool
rental_price_per_day	Monetary	Shortcut to first pricing rule's price
late_fee_per_day	Monetary	Daily late fee from first pricing rule
brand / model_name	Char	Manufacturer brand and model for reference
purchase_price	Monetary	Purchase cost (for ROI tracking)
total_revenue	Monetary (computed)	Sum of revenue from all completed rentals
product_id	Many2one	Linked product.product for inventory integration
location	Char	Physical storage location in warehouse

2.2 rental.order — Rental Order

The central document of the module. A Rental Order tracks the full lifecycle of a customer rental from draft through check-out, check-in, and final invoicing.

Order States

State	Meaning
Draft	Order created but not yet confirmed
Confirmed	Approved and scheduled for check-out

Checked Out	Tools physically handed to customer
Checked In	Tools returned; pending final calculation
Done	Order closed and totals finalized
Invoiced	Invoice generated and linked
Cancelled	Order cancelled; tools released

Key Fields — Header

Field	Type	Description
name	Char	Auto-generated rental reference (e.g. RNT/2025/001)
partner_id	Many2one	Customer (res.partner, individual only)
date_order	Date	Order creation date
date_planned_checkout	Date	Scheduled check-out date
date_planned_checkin	Date	Scheduled return date
date_checkout	Datetime	Actual check-out timestamp
date_checkin	Datetime	Actual return timestamp
rental_period_type	Selection	Billing period: Daily / Weekly / Monthly
rental_duration	Float	Planned number of billing periods
actual_duration	Float (computed)	Real duration based on checkout/checkin timestamps

Key Fields — Financials

Field	Type	Description
subtotal	Monetary	Sum of all line rental costs
tax_total	Monetary	Total tax amount across all lines
late_fee	Monetary	Total late fees from all lines
damage_charges	Monetary	Manual damage charges added at check-in
discount_amount	Monetary	Authorized discount deducted from total
advance_amount	Monetary	Advance collected at check-out
deposit_amount	Monetary	Security deposit held
payment_status	Selection	Paid / Unpaid
payment_method	Selection	Cash / Card / Bank / Credit
customer_rating	Selection	Perfect / Very Good / Good / Fair / Poor / Skipped

2.3 rental.order.line — Order Line

Each line on a rental order represents one tool (or product) being rented. Lines track individual quantities, durations, pricing, condition, and return status.

Field	Type	Description
tool_id	Many2one	The rental tool being rented
product_id	Many2one	Linked inventory product (auto-populated)
quantity	Float	Quantity checked out
returned_qty	Float	Quantity already returned
pending_qty	Float (computed)	quantity minus returned_qty
period_type	Selection	Day / Week / Month for this line
planned_duration	Float	Number of periods planned
actual_duration	Float (computed)	Real duration from timestamps
unit_price	Monetary	Price per period for this tool
rental_cost	Monetary (computed)	unit_price × actual_duration × quantity
late_fee_amount	Monetary	Late fee applied to this line
tax_percentage / tax_amount	Float / Monetary	Tax rate and computed tax amount
checkout_condition	Selection	Tool condition at check-out: Excellent to Damaged
checkin_condition	Selection	Tool condition recorded at return
is_partial_return	Boolean	Flagged when returned as part of a partial return

2.4 rental.pricing — Pricing Rules

Pricing rules define how rental fees are calculated. Multiple rules can be set per tool or category, with min/max duration bands for tiered pricing.

Field	Type	Description
tool_id	Many2one	Tool this rule applies to (optional)
category_id	Many2one	Category-level rule (applies to all tools in category)
period_type	Selection	Per Day / Per Week / Per Month / Fixed Price
price	Monetary	Rental price for this period
min_duration	Float	Minimum rental duration for this rule to apply

max_duration	Float	Maximum duration (0 = no limit)
late_fee_per_day	Monetary	Daily penalty for overdue return
is_primary_pricing	Boolean	Representative rule shown in the pricing list view

2.5 rental.expense — Business Expense

A standalone expense model scoped to the rental business. It mirrors Odoo's hr.expense pattern without depending on the HR module, providing a familiar expense workflow.

Field	Type	Description
name	Char	Short description of the expense (required)
date	Date	Expense date
user_id	Many2one	Who incurred the expense
category_id	Many2one	Configurable category (rental.expense.category)
quantity / unit_price	Float / Monetary	Quantity and unit cost
total_amount	Monetary (computed)	quantity × unit_price
payment_mode	Selection	Employee (to reimburse) / Company
payment_method	Selection	Cash / Card / Bank / Credit
rental_order_id	Many2one	Optional link to a specific rental order
state	Selection	Draft / Submitted / Approved / Refused / Posted
receipt_image	Binary	Attached receipt photo

3. Rental Workflow

The rental lifecycle follows a structured flow designed to minimize errors and maintain a clear audit trail at each stage.

Draft → Confirmed → Checked Out → Checked In → Done → Invoiced

(Cancelled is possible from Draft, Confirmed, or Checked Out)

3.1 Step-by-Step Flow

Step 1 — Create Rental Order (Draft)

- Select the customer from the partner list.
- Set the planned check-out and check-in dates.
- Choose a billing period: Daily, Weekly, or Monthly.
- Add tools to the rental lines. Pricing is auto-loaded from pricing rules.
- Optionally record an advance payment amount.

Step 2 — Confirm Order

- Validate the order to move it to Confirmed state.
- System checks tool availability against current checked-out orders.
- Confirmation locks the rental period and notifies via chatter.

Step 3 — Check-Out

- Launch the Check-Out Wizard from the order.
- Record the actual check-out date and time.
- Capture pre-rental tool condition per line (Excellent / Good / Fair / Poor / Damaged).
- Collect advance payment and record payment method.
- Attach checkout photos and customer signature via the camera widget.
- Wizard generates the Checkout Invoice PDF.
- Order state moves to Checked Out; tool availability reduces.

Step 4 — Partial Return (Optional)

- Use the Partial Return action to return a subset of tools mid-rental.
- Returned lines are flagged with returned_qty and partial_return_date.
- Remaining tools continue on the original order.

Step 5 — Check-In

- Launch the Check-In Wizard when tools are returned.
- System calculates actual duration (checkout timestamp to checkin timestamp).
- Late fees are computed automatically if return is after planned date.
- Record per-line tool condition at return and note any damage.
- Add damage charges and finalize the discount (if any).

- Collect remaining payment and record customer rating.
- Attach check-in photos and customer signature.
- Check-In Invoice PDF is generated.
- Order state moves to Checked In; tool availability restored.

Step 6 — Done & Invoice

- Confirm final amounts and mark order as Done.
- Generate the Rental Agreement PDF if required.
- Create the Odoo invoice from the order for accounting integration.
- Order moves to Invoiced state once the invoice is validated.

3.2 Discount Authorization Workflow

- A discount can only be applied via the Discount Wizard.
- The wizard requires the authorizer's name and a signature or photo.
- Once applied, discount_authorized_by, discount_auth_signature, and discount_applied_date are recorded.
- A timesheet entry is auto-created for the discount event.
- The discount deducts from the order total before payment calculation.

3.3 Tax Application

- Taxes are not applied automatically. Use the Tax Wizard button on the order.
- The wizard allows setting a tax percentage per line.
- Tax amount is calculated as: tax_percentage / 100 × rental_cost.
- Totals are recomputed instantly and displayed on the order form.

4. Pricing Configuration

The pricing engine is flexible and supports per-tool rules, category-level rules, duration-based tiers, and late fees.

4.1 Pricing Rule Priority

- Tool-specific rules always take priority over category-level rules.
- When multiple rules match a duration, the rule with min_duration closest to (but not exceeding) the actual duration is used.
- A rule with max_duration = 0 means no upper limit.

4.2 Period Types

Period Type	Day Multiplier	Use Case
Per Day	1	Short-term rentals; common for power tools
Per Week	7	Weekly project rentals; discounted daily rate
Per Month	30	Long-term rentals; best value for extended use
Fixed Price	N/A	One-time flat fee regardless of duration

4.3 Late Fee Calculation

Late fees are computed when the actual check-in date exceeds the planned check-in date:

```
Late Days  =  (date_checkin – date_planned_checkin) in days
Late Fee   =  late_fee_per_day  ×  Late Days  ×  quantity
```

5. Reports & Documents

5.1 PDF Reports

Report	Description
Rental Agreement	Full rental contract with terms, tools, pricing, and signatures
Checkout Invoice	Printed at checkout: tools, advance, deposit, and payment method
Check-In Invoice	Printed at return: final amounts, late fees, damage, discount
Partial Return Invoice	Issued when some tools are returned mid-rental
Tool Report PDF	Per-tool rental history and revenue summary

5.2 Dashboard & Analytics

- Live KPI tiles: Total Rentals, Active Rentals, Revenue, Overdue Orders.
- Order Reports dashboard: filter by date, customer, and status.
- Discount Details dashboard: audit all applied discounts.
- Tax Details dashboard: tax collected by period.
- Sales Report: revenue trend by billing period.
- Pricing Rules dashboard: quick view of all tool pricing.
- Categories dashboard: tools grouped by category with counts.

5.3 Excel Export

- The rental report wizard allows exporting order data to Excel (.xlsx).
- Requires the xlsxwriter Python library (optional dependency).
- Export includes all order lines, durations, costs, and customer info.

6. Security & Access Control

6.1 User Groups

Group	Permissions
Rental User	Create and edit own rental orders, view tools and pricing
Rental Manager	Full access: configure pricing, manage tools, approve discounts, access reports

6.2 Multi-Company Rules

- Each rental order, tool, pricing rule, and expense is scoped to a company.
- Multi-company record rules prevent cross-company data access.
- Users in multiple companies see only their company's records unless explicitly shared.

6.3 Discount Authorization

- Discounts require a named authorizer and a signature or photo.
- The authorization timestamp is immutably recorded on the order.
- Only Rental Managers can apply discounts above a configured threshold.

7. Installation & Configuration

7.1 Prerequisites

- Odoo 19.0 (Community or Enterprise)
- Python 3.10+ with standard Odoo dependencies
- xlsxwriter (pip install xlsxwriter) — optional, for Excel export

7.2 Installation Steps

- Copy the tools_rental_management folder into your Odoo addons directory.
- Restart the Odoo server: sudo systemctl restart odoo
- Go to Apps, search for 'Tools Rental Management', and click Install.
- The post_init_hook will automatically sync existing product categories to rental categories.

7.3 Initial Configuration

- Navigate to Tools Rental > Configuration > Tool Categories to create categories.
- Go to Tools Rental > Configuration > Expense Categories to set up expense types.
- Add tools under Tools Rental > Tools & Equipment.
- Define pricing rules on each tool or at the category level.
- Assign rental user groups to staff members as appropriate.

7.4 Scheduled Actions

Cron Job: Check Late Returns

Frequency: Daily

Purpose: Marks orders as overdue and notifies managers when checked-out tools are not returned by the planned date.

8. Technical Notes

8.1 Product Synchronization

The rental.tool model maintains a bidirectional sync with product.product:

- When a product is linked to a tool, tool fields (name, code, barcode, image, category) are auto-populated from the product.
- When tool fields are updated, changes propagate to the linked product automatically.
- The _syncing_rental_pricing context flag prevents infinite sync loops.
- Deleting a tool also deletes its linked product (cascade unlink).

8.2 Timesheet / Audit Trail

- Every significant event on a rental order (checkout, checkin, discount, damage, tax) creates a rental.timesheet entry.
- Timesheets store the action type, amount, user, and timestamp.
- This provides a complete financial audit trail for each order.

8.3 Camera Widget

- A custom Owl/JS camera widget (camera_widget.js) enables photo capture directly in the browser.
- Captured images are stored as base64 binaries in rental.order.media records.
- Separate media collections exist for checkout, checkin, and discount authorization.

8.4 Barcode Support

- Each tool can have a unique barcode field.
- Barcode uniqueness is enforced by a SQL constraint.
- The barcode can be scanned via any connected USB or Bluetooth scanner in the browser.

8.5 Migration Scripts

- Version 19.0.2.1.0 includes pre-migrate and post-migrate scripts.
- pre-migrate.py: Backs up or transforms data before schema changes.
- post-migrate.py: Re-establishes relationships and computes initial values after migration.

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This document was generated from the tools_rental_management module source code.